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PI Name:	Dynlacht, Joseph Robert Ph.D.		
Project Title:	Effects of Estrogen on Cataract Induction After Exposure to High LET Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	46202-5121	**Congressional District:**	7
Comments:			
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Task Description:	Several studies have demonstrated that there is an increased cataratogenic potential in animals and humans exposed to high linear energy transfer (LET) radiation compared to low LET radiation. Since the relative biological effectiveness (RBE) of heavy energetic charged (HZE) particles for lens opacification is high, these studies imply a high cataract risk for astronauts involved in long-duration space missions. During a manned mission to Mars, ocular doses in excess of ~0.1 Gy of HZE particles could be accumulated. Though not directly life-threatening, radiation-induced cataractogenesis represents a potentially serious sequelae which could necessitate surgical intervention in the years after, or perhaps even during the mission in some individuals (see below). While the cellular and molecular mechanism(s) of radiation-induced cataractogenesis have not been clearly elucidated, damage to DNA at the time of exposure and subsequent proliferation of the radiosensitive cells in the germinative zone of the lens epithelium likely play a role in the process. Using a rat model, we have recently accumulated data which indicate that estrogen increases or reduces the latent period and may modulate the incidence and severity of cataracts induced by low LET radiation, depending upon when the estrogen is administered. High estrogen levels are induced in non-pregnant women using oral contraceptives, or in post-menopausal women on estrogen replacement therapy. Therefore, sub-groups of female astronauts may be at an increased risk for developing cataracts that are more severe or occur with a more rapid onset. The interaction between estrogen and high LET radiation has never been studied before in the context of cataractogenesis. Therefore, a mechanistic approach characterizing the modulating effect of estrogen on high LET radiation-induced cataractogenesis is proposed. The underlying hypothesis of this proposal is that estrogen alters the latent period for high LET radiation-induced cataractogenesis, and, depending upon when the it is administered, modulates severity and incidence of cataracts. Estrogens regulate several proteins involved in cell cycle control and apoptosis, and it's metabolism results in the production of free radicals which may be genotoxic and mutagenic to mammalian cells. Thus, a novel hypothesis to be tested in the proposed studies is that estrogen alters cell cycle regulation, DNA double strand break (DSB) induction or repair, and proliferation in lens cells exposed to HZE particles (1 GeV/amu iron ion beams). The first Specific Aim shall be to define the dose-time interactions of high LET radiation (vs. low LET radiation) and estradiol to better understand the mechanism of estrogen action. The RBE for cataractogenesis in estrogen-treated animals exposed to1 GeV iron beams shall be determined using 60-Co gamma rays as a standard. The second specific aim shall be to determine the mechanism of estrogen-modulation of radiation-induced cataracts. Animal and cell culture studies will be performed to determine the effects of estrogen on gene expression, cell proliferation, apoptosis, and the induction and repair of radiation-induced DSBs. Slit-lamp biomicroscopy will be used to measure lens opacification in exposed animals. Successful completion of these studies will lead to a greater understanding of the modulation of space radiation effects by hormones and novel intervention strategies for long-duration space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results obtained from this project may provide greater insight into the potential effects of hormones on normal tissues after radiotherapy treatments with high LET particles.
Task Progress:	This report summarizes progress during the second full year of support. Work continued to be predominantly focused on the first Specific Aim of the project, which is to determine whether estrogen (17-beta-estradiol, E2) alters cataractogenesis induced by high LET radiation. During the second year of funding, we completed our first two "runs" at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL); beam time was requested in the Fall 2006 and Spring 2007 runs. Prior to our first run, we met with and subsequently corresponded with NSRL physicists and an engineer to design an animal holder and collimator set-up that would meet our specific needs. This device enabled us to irradiate the right eyes of up to 3 rats simultaneously, while sparing the contralateral eye and most other normal tissues. This resulted in other structures receiving a dose of less than 5% of the dose to the target eye. Over two hundred rats have been irradiated in this manner, and the holder/collimator device has been successful in eliminating the morbidity and mortality that would normally be encountered if the whole head of each rat were irradiated. Beam time requirements were also reduced three-fold. Prior to the runs at NSRL, an experiment involving sparsely ionizing (or low LET) radiation (60-Co gamma-rays) was initiated to assess gender-based differences in cataractogenesis after exposure to low LET radiation. The experiment would also allow us to determine whether estrogen would modulate radiation cataractogenesis. Placebo-implanted or E2-implanted 56-day old male rats and ovary-intact placebo-implanted 56-day old female rats were irradiated at Indiana University with 10 Gy using the Leksell Gamma Knife to irradiate only the right eye of each test animal with 60-Co gamma-rays. Slit-lamp biomicroscopy was used to measure lens opacification in exposed animals. Severity of cataracts was then measured using a scoring system described by Dynlacht et al. (Radiat Res. 165, 9-15, 2006). Opacification of the anterior and posterior subcapsular lenses of irradiated and control (contralateral) eyes was measured every 2-4 weeks. Data accrual for this first experiment is nearly complete, and two conclusions are evident after preliminary statistical analysis of the results. There appears to be no difference in the severity and progression of radiation-induced cataracts between male placebo- or E2-implanted rats, and female rats. However, there appears to be a significant difference in the incidence of anterior and posterior subcapsullar cataracts between male and female placebo-implanted rats. Thus, there appears to be some differences attributed to gender, at least in the case of cataracts induced by sparsely ionizing radiation. Interestingly, the gender differences do not appear to be due to a presence or a lack of estrogen since male rats with placebo implants showed no difference in cataractogenic potential when compared to male rats with E2 implants which were exposed to low LET radiation. Similar experiments were initiated during the Fall 2006 run at NSRL that will allow us to determine whether a similar response is obtained after E2- or placebo-implanted animals are exposed to high LET radiation. That is, we shall test the hypotheses that potential gender-based differences in incidence and latent period for cataractogenesis after exposure of rats to high LET radiation are related to steroid hormones. Placebo-implanted or estrogen-implanted 56-day old male rats and ovary-intact placebo-implanted 56-day old female rats were irradiated with 1 Gy of a 600 MeV/amu 56-Fe beam during the Fall 2006 run. Appropriately treated sham-irradiated control rats were also sent to the NSRL facility. Also, during the Fall 2006 run and continuing through the Spring 2007 run, a radiation dose response experiment was initiated at NSRL to determine whether modulation of cataractogenesis by E2 after exposure to high LET radiation is dependent upon the radiation dose. Ovariectomized or intact female rats, or ovariectomized rats that received

	estrogen-implants were irradiated with 0.1 to 2 Gy of 600 MeV/amu 56Fe ions. Data accrual has only recently begun for animals irradiated at NSRL during these runs. When data accrual for these experiments is complete, the issue of gender differences in radiation cataractogenesis will be resolved, and may have important implications in regard to the selection of crews for extended manned space flight.
Bibliography Type:	Description: (Last Updated: 07/17/2025)
Abstracts for Journals and Proceedings	Dynlacht JR, Henderson MA, Valluri S, DesRosiers C, Lopez JT, Batuello CN, Caperell-Grant A, Mendonca MS, Bigsby RM. "Gender-Related Differences in Cataractogenesis After Exposure to Low or High LET Radiation" Conference Center, League City, TX, February 12-14, 2007. 2007 NASA Human Research Program Investigators' Workshop, February 2007. , Feb-2007